Database Design Core V1

Flow:

Requirements → Core Domain/Data Model → ER Diagram → PostgreSQL → ORM → Backend

Core Entity Identification

#	Entity	Why it is needed
1	**Administrator**	The system's authenticated user who manages the business data.
2	**Business**	Stores the freelancer/business information used throughout the system and on invoices.
3	**Customer**	Represents the individual/company receiving contracts and invoices.
4	**Contract**	Represents an agreement between the business and a customer and can be associated with invoices.
5	**Invoice**	Core financial document managed by the system.
6	**Invoice Item**	Stores the individual products/services, quantities, and unit prices belonging to an invoice.
7	**Payment**	Records payments made against invoices and maintains payment history.
8	**Reminder**	Represents scheduled customer invoice reminders and their sending status/history.
9	**Invoice Delivery**	Records invoice email delivery attempts/status required by FR-037.
10	**Administrator Notification**	Represents critical notifications sent to the administrator.
11	**Report**	Represents/generated analytical reporting functionality, if reports need persistent records; otherwise reports

#	Entity	Why it is needed
		can be generated dynamically.
12	**System Settings**	Stores configurable business/system settings such as numbering, currency, tax, reminder schedule, and notification preferences/templates.

Entity-to-Requirement Traceability

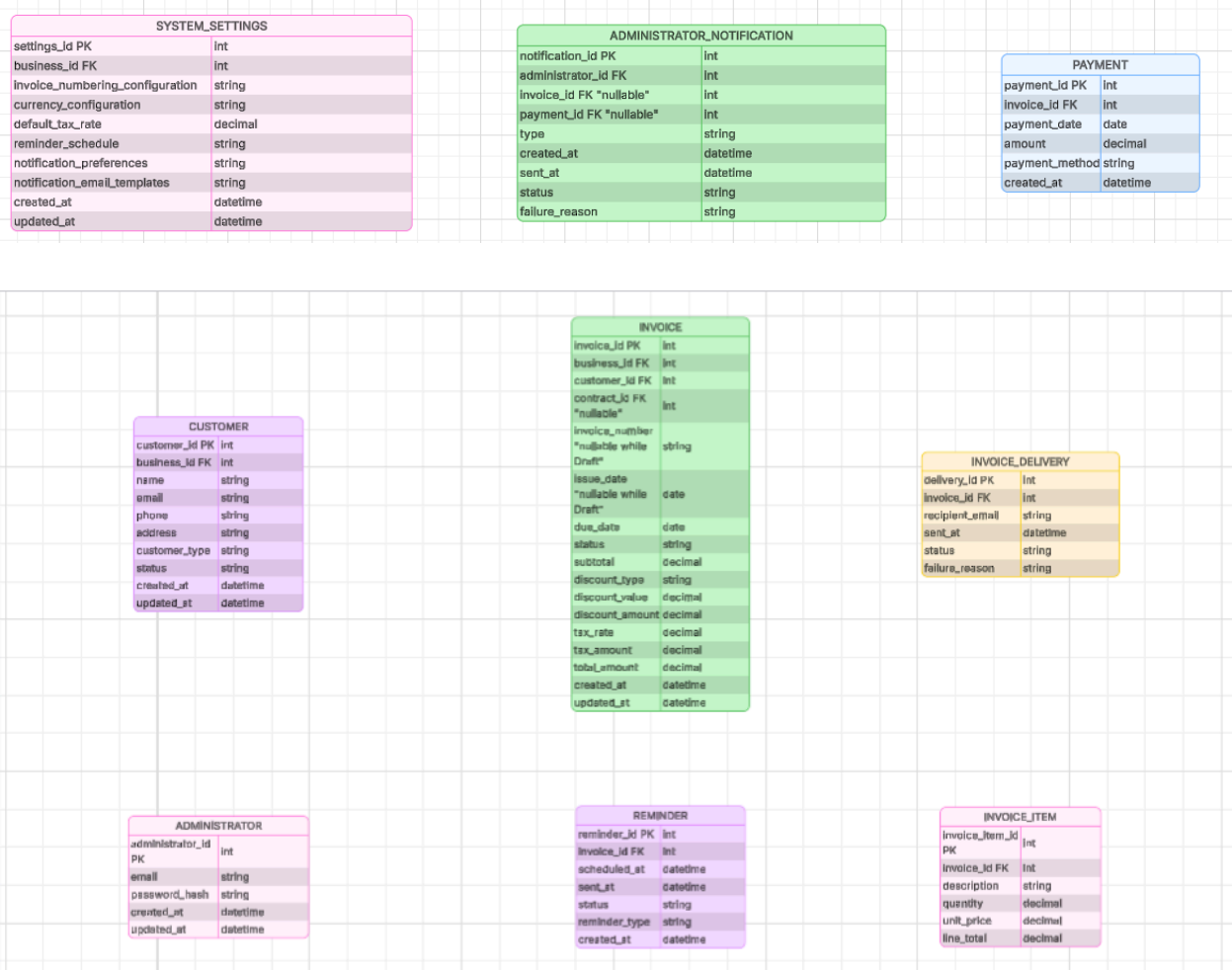
Entity	Main requirements it supports	Why the entity is needed
Administrator	Authentication, dashboard, management functions, settings	Stores the authenticated system user/administrator and their account information.
Business	Business profile/settings, invoice generation, invoice PDF	Stores the freelancer/business information that is displayed on invoices and used by the system.
Customer	Customer Management, Invoice Management, Contract Management, Reports	Stores the customers/clients who receive contracts and invoices.
Contract	Contract Management, invoice-contract association	Stores agreements between the business and customers and provides the contract reference for related invoices.
Invoice	Invoice Management, PDF generation, delivery, payments, reminders, reports	Core financial entity. Stores the invoice itself, its dates, totals, status, and customer/contract association.
Invoice Item	Invoice creation, calculations, PDF	Stores the individual services/items, quantities, and unit prices that make up an invoice.

Entity	Main requirements it supports	Why the entity is needed
Payment	Payment Management, invoice balance, reports	Stores every recorded payment made against an invoice and preserves payment history.
Reminder	Reminder Management	Stores scheduled customer reminders and their execution/status information.
Invoice Delivery	Invoice delivery / FR-037	Stores the history/status of invoice email delivery attempts.
Administrator Notification	Critical administrator notifications	Stores critical notification events and their sending/status information.
System Settings	Settings Management	Stores configurable system/business settings such as invoice numbering, currency, tax, reminder configuration, and notification preferences/templates.

Entity Attributes & Responsibilities

Note the below is not ER-Diagram, Only proposed attributes for the Entities





Relationship & Cardinality Analysis

1. An administrator **manages** a business. (Administrator 1 : 1 Business)

- Each Administrator manages exactly one Business.
- Each Business is managed by exactly one Administrator.

2. Business **manages** Customer — freelancer can work with many clients (Business 1 : N Customer)

- One Business can have many Customers.
- Each Customer belongs to exactly one Business.

3. Business **manages** Contract (Business 1 : N Contract)
- One Business can manage many Contracts.
- Each Contract belongs to one Business.
4. Customer **has** Contract (Customer 1 : N Contract)
- One Customer can have multiple Contracts.
- Each Contract belongs to exactly one Customer.
5. Business **issues** Invoice (Business 1 : N Invoice)
- One Business can issue many Invoices.
- Each Invoice belongs to one Business.
6. Customer **receives** Invoice (Customer 1 : N Invoice)
- One Customer can receive many Invoices.
- Each Invoice belongs to exactly one Customer.
7. Contract and Invoice (Contract 1 : N Invoice)
- A Contract can have zero or many related Invoices.
- An Invoice can optionally belong to one Contract.
- An Invoice does not have to have a Contract.
8. Invoice **contains** Invoice Items (Invoice 1 : N InvoiceItem)
- One Invoice contains one or more Invoice Items.
- Each Invoice Item belongs to exactly one Invoice.
9. Invoice **records** Payment (Invoice 1 : N Payment)
- An Invoice can have zero or many Payments.
- Each Payment belongs to exactly one Invoice.
10. Invoice **schedules** Reminder (Invoice 1 : N Reminder)
- An Invoice can have zero or many Reminders.
- Each Reminder belongs to exactly one Invoice.
11. Invoice **delivers** Invoice Delivery (Invoice 1 : N InvoiceDelivery)

- An Invoice can have zero or many delivery records.
- Each delivery record belongs to exactly one Invoice.

12. Administrator **receives** Administrator Notification (Administrator 1 : N AdministratorNotification)

- One Administrator can receive many notifications.
- Each notification belongs to one Administrator.

13. Invoice **references** Administrator Notification (Invoice 1 : N AdministratorNotification)

An administrator notification may be generated because of an invoice event, such as:

- invoice becomes overdue
- invoice becomes significantly overdue

But not every notification necessarily needs to reference an invoice.

14. Payment and Administrator Notification (Payment 1 : N AdministratorNotification)

The notification can reference the payment that triggered it.

15. System settings **configures** Business(Business 1 : 1 SystemSettings)

- Each Business has one configuration record.
- Each SystemSettings record belongs to one Business.

ER-Diagram V1:

generated by dbDiagram.io

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